

Karthikeya Chetharajula

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Professional Summary

Software Developer with 3+ years of experience delivering secure, scalable web applications in C#/.NET (ASP.NET Core, MVC, .NET 6/7, Web API) and Angular (up to 17), with complementary Python (Django/Flask). Builds reliable REST APIs and data workflows across SQL Server and MongoDB, automates CI/CD, and improves performance and security. Hands-on with Generative AI (RAG, embeddings, prompt engineering, tool/function calling) to accelerate search and support workflows.

Skills

Languages: C#, Python, JavaScript, HTML5, CSS3, Java

Web/Backend: ASP.NET Core 6/7, ASP.NET MVC 5, REST APIs, WCF, Swagger/OpenAPI

Frontend: Angular (up to 17), React, Kendo UI, AG Grid, Angular Material

Databases: SQL Server (incl. Spatial), MySQL, Oracle, MongoDB; T-SQL, stored procedures, indexing, query tuning

Cloud/DevOps: Azure (App Services, Functions, Azure SQL), AWS (Lambda, EC2, S3, RDS, Step Functions); Azure DevOps, GitHub Actions, Jenkins

Tools: Visual Studio, VS Code, SSMS, Postman, SoapUI, Swagger, Git, Azure DevOps, JIRA

Design Patterns: MVC, Repository, Factory, Dependency Injection (DI)

Testing/GenAI: xUnit, NUnit, MSTest, PyTest, Selenium; RAG, embeddings, evaluation/guardrails

Methodologies/DevOps: Agile/Scrum

Professional Experience

Software Developer

Apr 2024 – Present | Lenexa, KS

Clinical Reference Laboratory (CRL)

- Build and enhance features in ASP.NET (WebForms/MVC) and .NET 6/7, improving specimen/test processing throughput by 30%.
- Develop REST APIs and integrate Angular front-ends (Kendo UI) to enable secure, near real-time test status, results, and audit visibility.
- Author and tune T-SQL and stored procedures in SQL Server 2022 to streamline reconciliation, reporting, and chain-of-custody queries.
- Modernize legacy VB.NET/C# modules toward .NET Core services to improve maintainability and runtime performance.
- Implement RabbitMQ-based async messaging to improve responsiveness and scalability under load.
- Deploy Azure App Services/Functions/Azure SQL with autoscaling; achieved 99.9% uptime across high-volume workflows.
- Automate API testing with Postman/Swagger/xUnit and integrate quality checks into CI/CD (Azure DevOps/Jenkins).

Software Developer

Aug 2023 – Mar 2024 | Tampa, FL

SiteZeus

- Built .NET Core/C# Web API services for branch/ATM scoring, improving p95 latency by 30%.
- Implemented geospatial analytics (drive-time isochrones, catchment areas) using SQL Server Spatial, delivering 2–3x faster queries.
- Developed Angular 17 dashboards (AG Grid/Angular Material) with what-if scenarios, reducing analyst time-to-decision by 20%.
- Hardened APIs with OAuth2/JWT, RBAC, and audit logging to align with banking data-protection requirements.
- Productionized releases with CI/CD regression suites (SoapUI/Postman) and blue-green deployments in Azure DevOps/Jenkins.

Associate Software Engineer
Sonata Software

Jul 2020 – Dec 2021 | India

- Delivered ASP.NET MVC applications with Razor, Entity Framework, and WCF for real-time transaction capture and compliant reporting.
- Optimized SQL Server performance via indexing, stored procedures, and query tuning for high-volume reconciliation and reporting.
- Built .NET Core REST services and Python (Django/Flask) workers for batch/ETL processing to improve throughput.
- Deployed AWS components (Lambda, Step Functions, EC2) with autoscaling and workload offload to serverless, reducing operational cost by 20%.
- Implemented responsive UI features (JavaScript/AJAX, HTML, CSS) and automated tests (xUnit/NUnit, Selenium), achieving 95% coverage.

Selected Projects (GenAI)

- RAG Knowledge Assistant (C#/.NET 7, Angular): built embeddings + retrieval pipeline over documents; produced grounded answers with citations; added PII redaction and quality checks.
- Structured Extraction Helper (Python/Flask): converted unstructured notes into validated JSON schemas using LLM prompts; implemented retries, format validation, and automated tests.

Education

M.S., Computer Science — Missouri State University, Springfield, MO	2022–2023
B.S., Computer Science — Kakatiya University, Warangal, India	2016–2020

Certificates

Microsoft Certified Azure Fundamentals (AZ-900)
Certification in Data Science — Missouri State University